



Establishment of institutional Diagnostic Reference Levels for Computed Tomography examination at the University of Ghana Medical Centre

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Abstract

Introduction Diagnostic Reference Levels are important indicators for monitoring and optimizing protocol and procedure in medical imaging between facilities and equipment. This helps to evaluate whether, in routine clinical conditions, the median value obtained for a representative group of patients are within an agreed range from a specified procedure is unusually high or low for that procedure. This study aimed to propose Diagnostic Reference Levels for Computed Tomography examination of the most common routine examination of the head, chest and abdominal pelvis regions at the University of Ghana Medical Centre.

Methods The Diagnostic Reference Levels were determined based on the investigation of the most common routine examinations including head Computed Tomography examination with and without contrast, abdominopelvic Computed Tomography examination with and without contrast, and chest Computed Tomography examination without contrast. The study was based on two dose indicators; the volumetric Computed Tomography Dose Index and Dose-Length Product.

Results The estimated median distribution for head Computed Tomography with contrast for volumetric-Computed Tomography dose index and Dose-Length Product were **38.33 mGy** and **829.35 mGy.cm**, while without contrast, were **38.90 mGy** and **860.90 mGy.cm** respectively. For an abdominopelvic Computed Tomography examination with contrast, the estimated volumetric-Computed Tomography dose index and Dose-Length Product values were **40.19 mGy** and **2096.60 mGy.cm**. While without contrast, the estimated values were **14.65 mGy** and **800.40 mGy.cm** respectively. Additionally, for chest Computed Tomography examination, the estimated values were: **12.75 mGy** and **423.95 mGy.cm** for volumetric-Computed Tomography dose index and Dose-Length Product respectively. These median values represent the proposed diagnostic reference values of the head, chest and abdominal pelvis regions.

Conclusions The proposed Diagnostic Reference Level are comparable to recommended International Atomic Energy Agency and International Commission Radiation Protection Publication 135 and other regional published data by European Commission and Regional National Diagnostic Reference Level in Africa. These reference levels will serve as benchmarks to guide clinicians in optimizing radiation dose levels while ensuring accurate diagnostic image quality at the facility.

Keywords Diagnostic reference levels · Computed tomography dose index · Computed tomography · Radiation exposure · Dose-length product · Radiation protection

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1 Introduction

Computed Tomography (CT) plays a significant role in modern clinical medicine, by providing images of anatomical structures for accurate diagnosis and treat various medical conditions. However, the increasing use of CT procedures raises concerns about the associated radiation exposure and potential health risks to patients. To address these concerns, European Commission (EC) Directive on radiation

protection of 1999 (99/109/EURATOM) and International Commission on Radiological Protection (ICRP) in ICRP Publication 135 (initially Publication 73) provided some fundamental bases for the establishment of Diagnostic Reference Levels (DRL) for medical imaging [1–3]. The purpose of a DRL is to identify facilities where investigation of practices is not optimised in reference to a balance between image quality and patient dose. Good practice with regard to this optimisation process is to ensure that image quality remains commensurate with the clinical purpose of the examination in addition to significant application of patient dose reduction techniques.

The radiation dose received during CT examinations by patients is influenced by a number of factors including the equipment setting, procedure and the protocol used [4]. To ensure patient safety and optimize radiation dose levels, DRLs are established as valuable benchmarks values for monitoring and controlling radiation doses in CT examination without compromising image quality. DRLs are critical for monitoring and comparing radiation doses between different diagnostic imaging facilities and institutions but not to compare individual patient dose values. The goal of this comparison is to identify unusual high or low exposure values of dose indices, such as Computed Tomography Dose Index (CTDI) and Dose-Length Product (DLP), specific to a procedure under routine clinical conditions [2, 4, 5].

Additionally, DRL value are established by using the median distribution of these dose indicators observed in clinical practice within an institution, country or region [6]. However, local, regional or national DRL are established by using the 75th percentile of the various median distributions of individual institutions.

Furthermore, DRLs in clinical practice of CT rely on standardized indicators of CTDI and DLP for the most common procedure in a given facility mainly for optimizing doses. It is however important to note that CTDI and DLP are the output parameters of the CT scanner and do not provide estimates of patient dose [7]. Hence, the investigation here will only establish the median distribution of these dose parameters for single CT equipment at the facility to assist and promote patient safety and dose optimisation.

Additionally, several other studies have adopted this procedure in establishing and utilizing DRLs as part of dose optimisation procedure. For instance, in a recent study, four African countries [8] used the techniques of patient dose report were comparable with other internationally

published DRLs [2, 9, 10]. Other publications such as Ekpo et al. and Dawd et al. in Nigeria, Benmessaoud in Morocco and Salama et al. in Egypt established national DRL in Nigeria, Morocco, Egypt and Algeria on the most common routine procedure in CT imaging and this has significantly improved process of dose optimisation in Nigeria, Morocco, Egypt, Algeria [11–15]. The IAEA and ICRP has published comprehensively data on DRL which has become the source of information for developing countries and researcher. These organisations have recommended the use of DRL for optimisation in ICRP Publication 135.

Finally, the outcome of these research findings on the routine clinical procedure for head, chest, and abdominopelvic CT examination will provide base for effective radiation dose optimisation and future research guidelines on imaging practices in Ghana's medical community [16–20]. This study aimed to propose the establishment of DRLs for Computed Tomography examination of the most commonly routine examination of the various body regions at the University of Ghana Medical Center.

2 Methodology

The methodology consists of materials and methods.

Materials used included Philip CT Scanner (ingenuity 128 slice) at the facility which was manufactured in 2014 and installed in the radiology department in 2016. Details of the specifications of the CT system used at the facility is shown in Table 1. Head and Body Phantom with calibrated ion chamber. The raw data was obtained from the picture archiving and communication (PACS) which made available comprehensive patients dose reports and patient biodata. This enable patients dose reports to be retrieved from the PACS and the patient biodata from the Radiology Information system (RIS). Initially, the daily Quality Control (QC) was assessed and validated using Head and Body Phantom with calibrated ion chamber.

The methods used involve two processes:

The first was by performing QC on the equipment and compare the measured parameters with those at acceptance to ensure the machine was working properly. The routine technique factors mostly used at the facility for the three most commonly used procedure were validated during the QC procedure. These factors were Kilovolts Peak (kVp), milliamp per second (mAs), Pitch, and patient size. Which were measured as part of the day-to-day radiological procedure at the facility.

The second was retrospectively data were retrieval from the PACS using IAEA designed data collection form. These were validated as part of the pre data collection and QC

Table 1 Details of the specification of the CT scanner

Manufacturer	Philips
Model	Philips CT ingenuity 128 slice
Slices	128
Year of Manufacture/Installation	2014/2016
Reconstruction Algorithm	Filtered Back Projection

procedure. Additionally, before the data collection process, measurements were done for the assessment and calibration of the results of the measuring instruments and the CT machines.

Furthermore, the CTDI and DPL were estimated from the head and body phantom to determine its accuracy as compared to the recorded values from the dose report on the PAC system and the those displayed on the console. The measurements were done with a calibrated ion chamber (PTW DIADOSE Meter, ref. number T11035) and compared with the recorded values from the console and the dose report. This was to ensure accuracy of the recorded values on the PACs and the console.

2.1 Procedure for data collection

Data from patients who had undergone CT examination between 2018 and 2023 were collected using a structured questionnaire provided by the IAEA, as part of the new regional project (RAF/9/059) for the establishment of national diagnostic reference levels (NDRLs) in Africa. A total of 150 dose reports from the image data of selected adult patients undergoing CT scans of head, chest, and abdominal pelvis were collected and analysed, from which 128 met the selection criterion and were used in this study. Additionally, a routine head, chest, and abdominal pelvis CT examinations for patients more than 18 years old were collected.

Before the data collection, basic daily QC were performed on the CT scanners, including calibration check (Fig. 1) kVp accuracy, mA accuracy, exposure time, Half Value Layer (HVL) and image noise assessment in term of Signal to Noise Ratio (SNR) (Fig. 2). Additionally, as part of the QC a specific patient information for each scan, including patient gender, age, peak tube potential (kVp), effective current-time product (mAs), scan length (mm), anatomical scanning range, acquisition slice thickness, and the use of automatic exposure control (AEC) were also collected and checked for accuracy and reproducibility.

Furthermore, after the basic QC were performed, a dosimetry estimates in terms of volume-CTDI ($CTDI_{vol}$) and DLP were also done. The weighted-CTDI ($CTDI_w$) was used to estimate $CTDI_{vol}$. The $CTDI_w$ was measured using a 16 cm and 32 cm diameter PMMA phantom with calibrated pencil chamber as shown in Fig. 3. The pencil dosimeter was placed serially in the centre hole and the peripheral hole and Eq. (1) used to estimate the $CTDI_w$ as shown below.

$$CTDI_w = \frac{1}{3}CTDI_{100/Centre} + \frac{2}{3}CTDI_{100/Periphery} \quad (1)$$



Fig. 1 Calibration checks



Fig. 2 Set up for comprehensive quality control



Fig. 3 Setup for measuring weighted CTDI

Where the 100/centre subscript denotes, measurements made with a 100 mm pencil probe at the centre. While the 100/periphery subscript represents an average of the periphery measurements at four different locations around the phantom as shown in Fig. 2.

The Eq. 2 was then used to obtain the by dividing the measured $CTDI_W$ by the pitch factor (p).

$$CTDI_{Vol} = \frac{CTDI_W}{P} \quad (2)$$

DLP was estimated by multiplying the $CTDI_{vol}$ by the scan length (L) as shown in Eq. (3).

$$DLP = CTDI_V * Scan Length \quad (3)$$

The estimated QC values were compared with the commissioning and acceptance tests results of the equipment to ensure that the CT machine was performing optimally and were still within accepted range of deviations. This was significant to confirm the commonly used protocols in the facility.

The next process after the QC procedure was the data collection process, where a minimum of 20 adult patient dose report from the PACS (Picture Archiving and Communication System) were retrieved. The parameters of interest were the $CTDI_{VOL}$ and DLP from the structured dose report, including the procedure used, the applicable optimisation protocol used and the disease condition with each patient specific CT examination protocol.

The median values of the $CTDI_{VOL}$ and DLP were then estimated and compared with a typical dose level of published DRLs for similar practice, in order to provide a broad indication of the relative performance of the imaging technique. The estimated median distributions were calculated as the estimated DRL value for a specific procedure of the facility [2, 8].

Furthermore, Table 2 indicates ICRP suggested CTDI and DLP reference median distribution for establishing DRL in ICRP publication 135.

3 Results

The study examined the CTDI and proposed Diagnostic Reference Levels (DRLs) for the modality with and without contrast in CT examinations of the head, chest, and abdominopelvic. The data obtained allowed for a comparison of the dose values between the contrast and non-contrast scans. An average of 25 scans for each procedure were used to compute the CTDI and DLP values as recommended in ICRP publication 135 [2].

Table 2 IAEA/ICRP suggested CTDI and DLP reference median distribution

Examination	Phantom	$CTDI_w$ (mGy)	DLP (mGy.cm)
Routine head	Head	60	1050
Routine chest	Body	30	650
Routine Abdominal Pelvis	Body	35	780

Table 3 Characteristics of the study population

Examination Type	Age in years			Total number of patients
	Min	Max	Mean	
Head (Contrast)	27	88	57.58	26
Head (Non-contrast)	23	88	57	30
Abdominopelvic (Contrast)	27	90	55.5	20
Abdominopelvic (Non-Contrast)	21	77	45.6	20
Chest (Non-contrast)	23	93	55.06	32

Table 4 Established DRLs of $CTDI_{vol}$ for this study

	$CTDI$ (mGy)				
	Head (Contrast)	Head (non-contrast)	AP (contrast)	AP (non-contrast)	Chest
Mean	39.07	38.57	42.6	14.12	12.29
Median	38.33	38.9	40.19	14.65	12.75
SD	3.07	5.01	11.03	2.55	1.86
Range	26.1	26.1	36.91	10.1	7.6
Minimum	33.99	21.5	29.19	8.5	9.1
Maximum	45.3	47.6	66.1	18.6	16.7
25th	36.72	36.93	32.65	12.55	10.97
50th	38.33	38.9	40.19	14.65	12.75
75th	41.18	40.33	51.02	15.94	13.12

Table 3 is the characteristics of the study population based on age and total number of patients in this study. It also provides detailed examination type, that is Head (contrast and Non-Contrast), Abdominal Pelvis (contrast and Non-Contrast) Chest examinations in the study. A total of 128 patients were involve in the study with 56 Head, 40 abdominal-Pelvis and 32 chest examinations.

Furthermore, Table 4 shows the established DRLs for the respective CT examinations of the head contrast and non-contrast, abdominal pelvis contrast and non-contrast and chest based on real patient data in terms of $CTDI_{vol}$.

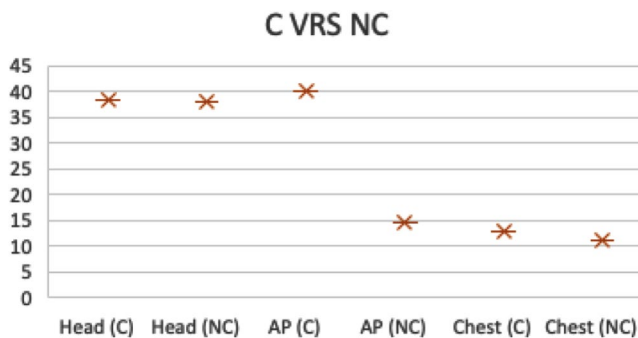
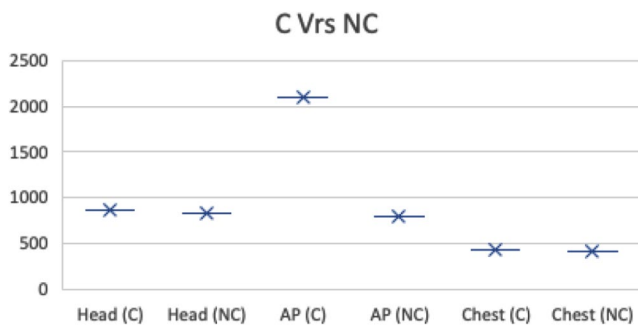
Additionally, Table 5 shows the established DRLs for the respective CT examinations of the head contrast and non-contrast, abdominal pelvis contrast and non-contrast and chest based on real patient data in terms of **DLP**.

Figure 4 shows the relationship between Contrast (C) and Non-Contrast (NC) for CTDI. This represent a pictorial

Table 5 Established DRLs of DLP for this study

	DLP (mGy-cm)				
	Head (Contrast)	Head (non-contrast)	AP (contrast)	AP (non-contrast)	Chest
Mean	826.93	869.69	2165.72	754.03	423.96
Median	829.35	860.9	2096.6	800.4	423.95
SD	67.97	144.51	485.61	136.62	69.69
Range	267.9	763.2	2145.3	479.8	385.5
Minimum	703.1	529	1399.4	460.9	253.8
Maximum	971	1292.2	3544.7	940.7	639.3
25th	771.15	805.7	1814.6	653.45	380.63
50th	829.35	860.9	2096.6	800.4	423.95
75th	876.33	900.03	2495.8	841.33	466.6

*AP = Abdominal Pelvis

**Fig. 4** Relationship between Contrast (C) and Non-Contrast (NC) for CTDI**Fig. 5** Relationship between Contrast (C) and Non-Contrast (NC) for DLP

trend observed between the procedure when contrast was administered and when it was used.

Figure 5 shows the relationship between Contrast (C) and Non-Contrast (NC) for DLP. This represent a pictorial trend observed between the procedure when contrast was administered and when it was used in terms of the dose parameters.

Table 6 represent the results of international published data from within our sub-region and other international literature. The results indicate similar trend when compared

with international published data within the sub-region but varied significantly from other national published data.

4 Discussions

Tables 4 and 5 shows the dose estimates for both the contrast and non-contrast examinations. The trend of the result indicates that, the CTDI values for the scans performed with contrast were significantly higher than those without contrast with an average variation of 200% +/- 12% in some cases. This difference indicates the additional radiation dose delivered when contrast is used to enhance the imaging process.

For example, in head CT scans, the median distribution for scans with contrast was found to be higher than those without contrast. This indicates that the addition of contrast agent resulted in a higher radiation dose to the patient. However, the difference between these were not higher with percentage variations of 8.03% and 0.53% respectively for CTDI and DLP values. A similar trend was observed in the abdominopelvic CT scans with percentage variations of 219.6% for the CTDI and 206.0% for the DLP with contrast. This appreciable increase from non-contrast to contrast based abdominopelvic examination can be explained by the fact that, for the latter, the scan is done in three phases that is pre-contrast, post contrast, and delay. Hence, these repeated processes exposed the patients to repeated scans resulting in higher radiation dose, as the application of contrast medium is accompanied by an increase in radiation absorption in the tissues which leads more absorbed dose to the patients.

The study established DRLs for both contrast and non-contrast CT examinations of the brain, chest, and abdominopelvic at the University of Ghana Medical Center. The findings indicates that the CTDI and DLP values obtained were comparable with reported benchmarks as presented in Table 6. These values represents the median values for the dose estimates of the dose distribution. The finding suggest that the Centre significantly implement the optimisation of radiation dose protocol based on the comparable results of the QC and the proposed estimated DRL values. Additionally, the results of the QC value comparable to the initial measurement done during the acceptance and commissioning test at the Center 6.7% +/-1.2%.

Additionally, the data retrieved from the standard dose report considered agenda weight variation of the dose indices as recommended in ICRP publication 135 and 103 [2, 3] However, analysis of the data shows that there was no significant difference between the gender and weight specific variation, hence have not been included in the presentation and discussions.

Table 6 Published international national DRLs

Country	Head		Chest		AP	
	CTDI (mGy)	DLP (mGy.cm)	CTDI (mGy)	DLP (mGy.cm)	CTDI (mGy)	DLP (mGy.cm)
Ethiopia ⁶ (2023)	53	1210	13	635	16	822
Morocco ¹³ (2022)	57.4	1020	12.3	632	10.9	714
Nigeria ¹¹ (2018)	61	1310	17	735	20	1486
Kenya ⁸ (2020)	61	1612	19	895	20	1845
Egypt ¹⁴ (2017)	30	1360	22	420	31	1325
Uganda ²² (2022)	56	1260	7.8	377	12.5	1418
Japan ²⁵ (2020)	77	1350	13	510	18	880
USA ²⁷ (2023)	57	1011	15	545	20	1004
Canada ²⁴ (2011)	82	1302	14	521	18	874
Germany ¹⁶ (2017)	60	850	10	350	15	700
Australia ²⁶ (2020)	52	880	10	390	13	600
UK ²³ (2022)	47	790	8.5	290	10	530
Regional DRLs (Ghana, Kenya, Namibia Senegal) ⁸ (2022)	60.9	1259	15.2	544	15.7	737
This study (Contrast/Non-Contrast)	38.33/38.90	829.35/860.90	12.75	423.95	40.19/14.65	2096.60/800.40
SD	3.07/5.01	67.97/144.51	1.86	69.69	11.03/2.55	485.610/136.62

Furthermore, the results of the data from the standard structured dose report were also comparable with less than 5% variations as published in three earlier publications in Ghana [17, 18]. The analysis of the results and the associated optimisation protocol and the recommended procedure provided by the manufacturer were comparable to various international standards and practices in medical imaging [2].

The comparison of CTDI and DLP values between contrast and non-contrast scans primarily during abdominopelvic scans underscores the need for careful consideration of the potential benefits and risks associated with the use of contrast agents as shown in Figs. 4 and 5. From the results, while contrast agents enhance diagnostic capabilities, they also contribute to an increased radiation dose. Therefore, the team of healthcare professionals must weigh the diagnostic benefits against the potential risks when deciding to use contrast in the abdominopelvic CT examinations.

The proposed DRLs served as reference benchmarks for evaluating the radiation dose levels. These were determined based on the median distribution among all patients for the modality. These DRLs guide healthcare professionals in optimizing radiation dose levels, aiming to strike a balance between diagnostic accuracy and potential radiation risks.

Furthermore, analysis of the data shows that, for brain scan, the CTDI values in this study, both with and without contrast, were within the reported range found in literature [16–30]. Similar observations were made with values from other countries, such as Nigeria, and Morocco, Egypt and Algeria [11–14], indicating consistency across different studies.

Additionally, analysis of the dose reports also shows that the DLP values for brain examination obtained in our study

were within the reported range in literature. These showed similar trend reported in estimated values from Uganda, and Morocco. Hence, both dose indices, CTDIvol and DLP shows significant consistency with regional benchmarks as indicated in Table 5 [12, 13, 25].

In terms of chest examination, the CTDI value in this study (13 mGy) was within the reported range as published in literature. These were in agreement with values from other countries, such as Ethiopia (13 mGy), Japan (13 mGy), and Morocco (12.3 mGy) as indicated in Table 5 [11, 12, 14, 15]. This suggest that radiation dose estimates in chest examination at the facility are in line with international practices.

Furthermore, similar trends were also observed with regards to abdominopelvic examination. The CTDIvol values obtained in this study for scans without contrast were within the range reported in Ethiopia, Japan, Uganda and Morocco with slight variation of less than 5% [6, 22, 25, 29]. While the USA and Egypt reported higher CTDIvol values of more 5% as indicated in Table 5, based on reference publications [11, 12, 14, 16, 17, 29, 30].

Similar trends were observed with regards to the analysis of the estimated DLP values. The DLP values reported in this study for head, chest, and abdominopelvic examination were generally consistent with the findings from UK, and Australia [19, 26], with variation of more than 5%. These variations emphasize the importance of region-specific DRLs that are comparable with the local context, equipment, and study population.

It is interesting to note that, as indicated in Table 5, the regional DRLs for Ghana, Kenya, Namibia, and Sudan [8] were within the range of the reported values in this study [8,

15–18]. This indicates that the proposed DRLs in this study are comparable well with regional benchmarks, providing a valuable contribution to the establishment of radiation dose reference levels (DRL) at the Centre.

Overall, the comparison of results across different countries within the sub-region shows similar trends in radiation dose levels for CT examinations. This may be attributed to similar practices and level of technology in the sub-region. However, there were significant variation of the estimated dose indices outside the sub-region, these variations can be attributed to differences in scanning protocols, institutional and equipment protocol, equipment type, study populations, regulatory practices and technology. Hence, local-specific DRLs, as done in this study, may ensure that radiation dose levels in CT imaging are optimized and tailored to the local context, enhancing patient safety and contributing to the overall improvement of medical imaging practices in Ghana.

In conclusion, the proposed DRLs were established based on the median distribution of the rounded CTDIvol and DLP values for each examination from standard structured dose report. Based on the protocols and procedure recommended in IAEA recommended harmonized protocol for Africa. Following these protocols and procedures, this study has established DRLs for CT examination of the brain, chest, and abdominopelvic regions at the University of Ghana Medical Center, contributing to the ongoing efforts to promote radiation safety in medical imaging and align with international best practices.

4.1 Limitations

It is important to acknowledge some limitations of this study. Firstly, the study is an institutional based study with a single CT equipment, hence is purely meant to establish institutional DRL. Additionally, the Centre is contributing data on the ongoing efforts to establish national DRL in various modalities including CT. Future studies encompassing a broader range of medical equipment and Centres may comprehensively support our findings and establish comprehensive guidelines for radiation dose optimisation in Ghana. Secondly, the study focused on a specific time period and patient population at the University of Ghana Medical Centre, and radiation dose levels may vary over time and across different patient demographics. A regular update of this is recommended by ICRP publication 135 to be repeated every 5 years [2].

5 Conclusion

The findings of this study demonstrates that the CTDI and DLP values obtained at the University of Ghana Medical Centre are comparable well with reported benchmarks from the literature, indicating that radiation dose levels employed in medical imaging at the University of Ghana Medical Centre are consistent with international standards and practices. These findings highlight the commitment of UGMC to patient dose optimisation protocol while delivering high-quality medical imaging services. The study provides a solid foundation but calls for further research and collaboration to expand the understanding of radiation dose optimisation in Ghana's healthcare facilities and establish standardized guidelines for the wider medical community.

5.1 Disclaimer

This article contributed significantly to the efforts in establishing national and regional diagnostic reference level in Ghana and Africa. The team contribute data, reference material, equipment and other resources to the establishment of national and regional diagnostic reference level in Ghana and Africa. All material used are duly reference in this article. The article is intended to contribute mainly to dose optimisation at UGMC and to provide knowledge to readers.

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Author contributions All the Authors were involved pre and post data collection and analysis during the study. The following specific activities were done by the Authors. Concept Note: Issahaku Shirazu, T Simon Mensah Amoh, Isaac Kwesi Acquah, George Nunoo, Theophilus. Sackey. Pre-data collection activities including application for ethical clearance: Issahaku Shirazu, Issahaku Shirazu, T Simon Mensah Amoh. Data collection and analysis: All Authors (Issahaku Shirazu, T Simon Mensah Amoh, Isaac Kwesi Acquah, George Nunoo, Theophilus. Sackey). Drafting of Text: Issahaku Shirazu, Issahaku Shirazu, T Simon Mensah Amoh, Isaac Kwesi Acquah, George Nunoo, Theophilus A. Sackey. Review of Text: Issahaku Shirazu, Isaac Kwesi Acquah and Theophilus A. Sackey.

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Data availability The data and materials for publication are all available upon request at any time.

Code availability Not applicable.

Declarations

Ethical approval Approval was given for the research by the participating facility and the ethical and Protocol Review Committee of the University of Ghana Medical Centre. Additionally, the protocol and the application of same for this study was granted ethical clearance by the Committees. The methods were performed in accordance with the guidelines and regulations as outline by the ethical and Protocol Review Committee of the University of Ghana Medical Centre.

Consent to participate Not Applicable.

Consent for publication Not Applicable.

Competing/conflict of interests Not Applicable.

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